Software Requirements Specification

for

“NeuraNote”

Intelligent Study Companion for Structured Learning

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Table of Contents

1. Introduction
   - 1.1 Purpose
   - 1.2 Document Conventions
   - 1.3 Intended Audience and Reading Suggestions
   - 1.4 Product Scope
   - 1.5 References

2. Overall Description
   - 2.1 Product Perspective
   - 2.2 Product Functions
   - 2.3 User Classes and Characteristics
   - 2.4 Operating Environment
   - 2.5 Design and Implementation Constraints
   - 2.6 User Documentation
   - 2.7 Assumptions and Dependencies

3. External Interface Requirements
   - 3.1 User Interfaces
   - 3.2 Hardware Interfaces
   - 3.3 Software Interfaces
   - 3.4 Communication Interfaces

4. System Features
   - 4.1 Login & Authentication
   - 4.2 Study Material Upload & Processing
   - 4.3 AI-Driven Note Generation
   - 4.4 Note Management
   - 4.5 Quiz & Active Learning
   - 4.6 Progress Tracking & Analytics
   - 4.7 To-Do List & Calendar Integration
   - 4.8 Semantic Search (Vector-Based Retrieval)
   - 4.9 Notification & Reminder System
   - 4.10 Administrative Management

5. Other Nonfunctional Requirements
   - 5.1 Performance Requirements
   - 5.2 Safety Requirements
   - 5.3 Security Requirements
   - 5.4 Software Quality Attributes
   - 5.5 Business Rules

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for **NeuraNote**, an AI-powered web and mobile application that helps learners to organize study materials, generate structured notes, and improve knowledge retention through active learning and productivity tools.

This document serves as a contract between stakeholders, developers, supervisors, and evaluators by clearly describing what the system will do and the constraints under which it must operate.

1.2 Document Conventions

This document follows the formatting and organizational standards established by the **IEEE Std 830-1998** for Software Requirements Specifications. The following conventions are used throughout this report:

- **Section Numbering:** All section and subsection numbers follow the standard IEEE structure (e.g., 1.0, 2.1, 3.2.1) to maintain a clear hierarchy of information.
- **Priority Levels:** Functional requirements are assigned a priority level to guide the development phases:

  High: Critical core features (e.g., Note Synthesis, Iterative Modification Loop).

  Medium: Important but non-critical features (e.g., Notification System, Calendar Sync).

  Low: Optional enhancements (e.g., UI customization, offline preview).
- **Requirement Identifiers:** Every specific requirement is labeled with a unique ID (e.g., **REQ-4.x.x**) for easy traceability during the testing and implementation phases.
- **Visual Notations:**

  UML Diagrams: Standard Unified Modeling Language (UML 2.0) notations are used for Use Case, Sequence, and Activity diagrams.

  Bold Text: Used to highlight key technical terms, database entities, or specific UI elements like the **User Input Ribbon**.

- **Terminology:**

  "Shall" or "Must": Indicates a mandatory requirement that the system is obligated to satisfy.

  "Should": Indicates a recommended but non-mandatory requirement.

- **Technical Terms:** Specialized acronyms such as **JWT** (JSON Web Token), **LLM** (Large Language Model), and **CRUD** (Create, Read, Update, Delete) are defined in the Glossary (Section 1.3).

1.3 Intended Audience and Reading Suggestions

- **Developers & Testers** – Focus on Sections 2 (Overall Description), 3 (External Interfaces), 4 (System Features), and 5 (Nonfunctional Requirements).

- **Supervisors & Client** – Focus on Sections 1 (Introduction), 2 (Overall Description), and 5.

- **UI/UX Designers** – Focus on Sections 2.2, 2.3, and 3.1.

- **Project Managers** – Focus on Sections 1, 2, 4, and 5 to understand scope, constraints, and high-level features.

1.4 Product Scope

NeuraNote provides a unified platform where learners can upload study materials, transform them into structured notes using AI, practice with quizzes, track progress, and manage study schedules through an integrated to-do list and calendar.

Core benefits include:

- Centralizing study content and tasks in a single system.
- Improving knowledge retention via active learning and spaced repetition.
- Saving time by automating note creation and quiz generation.
- Supporting consistent study habits with reminders and analytics.

1.5 References

- NeuraNote Interim Report / Project Description (Team SquadZero).

- **IEEE Computer Society**, "IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications," 1998.https://ieeexplore.ieee.org/document/720574

- **ISO/IEC**, "25010:2011 Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE)," 2011.https://www.iso.org/standard/35733.html

- **Active Recall and Spaced Repetition Research:** Educational frameworks supporting the efficacy of automated quiz and flashcard generation for knowledge retention.

  Laynor, G. (2023). "Spaced Repetition: A Strategy for Long-Term Retention." *Journal of the Medical Library Association*.

  (Significance: Specifically discusses how digital spaced repetition tools optimize learning time for medical and professional students.)

- **Google Calendar API Overview:** Official documentation for synchronizing study tasks and deadlines with external calendar services.

  Google Developers (2024). "Google Calendar API Overview: Managing Events and Reminders." *Google Developer Documentation*.

  (Significance: The official technical standard for synchronizing academic tasks with external calendars.)

2. Overall Description

2.1 Product Perspective

Neura Note is a standalone, cross-platform application utilizing a high-performance client-server architecture. The system is built on a modern stack designed for speed and academic reliability:

It consists of:

- **Frontend:** Built with **React** to provide a responsive, single-page application experience.
- **Backend:** Powered by **FastAPI**, leveraging asynchronous processing for efficient handling of AI requests and user data.
- **AI Orchestration:** Managed via **LangChain**, connecting the backend to external Large Language Models (LLMs).
- **Unified Data Layer:** Utilizing **PostgreSQL** as the primary database for structured records, metadata, and **semantic search**. Vector capabilities within PostgreSQL are used to manage embeddings for study content, ensuring a streamlined data architecture.
- **Cloud Storage:** Integration with **Cloudinary or AWS S3** for secure, scalable hosting of user-uploaded study materials (PDF/PPT).

The system interacts with:

- External LLM services (e.g., OpenAI) for note and quiz generation.
- External calendar services (e.g., Google Calendar) for schedule integration.

2.2 Product Functions

At a high level, NeuraNote provides the following functions

- User registration, authentication, and profile management.
- Uploading and processing study materials (PDF, DOCX, images, text).
- AI-driven generation of structured notes from materials and user prompts.
- Note editing, tagging, organizing into courses/subjects, and exporting.
- Quiz generation from notes and quiz execution with feedback.
- Progress tracking dashboards and analytics (scores, streaks, coverage).
- To-do list and study scheduling with calendar integration and reminders.

2.3 User Classes and Characteristics

Registered Users

This is the primary user category for Neura Note. These individuals have established a secure profile via the authentication module and possess Full Access to the following suite of features:

- Iterative Note Synthesis (The Synthesis Loop): Users can upload PDF and PowerPoint materials and engage in unlimited modification cycles using the "User Input Ribbon" until the generated study guide is perfected.
- Active Recall Generation: Unrestricted access to the automated generation of digital Flashcards and interactive Quizzes derived directly from their synthesized notes.
- Progress Analytics & Visualization: Full visibility of their learning curve through Time-Series Graphs, which plot recall accuracy and consistency against time.
- Task Management & Habit Tracker: Complete use of the Calendar View to schedule study sessions, set reminders, and manage academic deadlines to build a consistent learning habit.
- High-Fidelity Document Export: The ability to download their final, AI-refined notes as professionally formatted PDF documents for offline review.
- Personalized Study Dashboard: A secure, persistent workspace where all notes, materials, and historical performance data are stored and managed.

System Administrators

Manages user accounts, monitors system health, and handles configuration.
Requires advanced technical knowledge and access to admin tools.

2.4 Operating Environment

Neura Note is designed to operate seamlessly across modern digital infrastructures, ensuring accessibility for students and professionals in various learning environments.

- **Client Side:** The application is compatible with the latest two versions of all major **Modern Web Browsers**, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari. Responsive design ensures functionality across desktops, tablets, and mobile devices.

- **Server Side:**
- **Backend Framework:** The system operates on a high-performance **FastAPI** framework (Python), utilizing its asynchronous capabilities to handle real-time AI orchestration and high-concurrency user requests efficiently.
- **Unified Data Layer:** The application utilizes **PostgreSQL** as the single source of truth for all operational data. This includes relational data for user accounts, task metadata, and quiz results.
- **Semantic Search Engine:** To facilitate long-term study memory and context retrieval, the system leverages **PostgreSQL's integrated vector-processing capabilities** (e.g., pgvector). This allows for high-speed semantic search and embedding storage within the same database environment, ensuring data consistency and simplified maintenance.
- **Cloud Object Storage:** User-provided resources, such as PDF and PPT materials, are stored and managed via **Cloudinary or AWS S3**, ensuring scalable and secure file handling outside the primary database.
- **Network:** A stable internet connection is mandatory to facilitate real-time AI synthesis and document parsing.
    - **Security Protocol:** To protect sensitive academic data, all communication between the client and server must be conducted over **HTTPS** using TLS encryption.

2.5 Design and Implementation Constraints

The development and deployment of Neura Note are governed by specific technical and ethical boundaries to ensure system sustainability, architectural simplicity, and user trust:

- **Technology Stack Mandate:** The system must be developed using React for the frontend web interface and FastAPI for the backend infrastructure to leverage high-performance asynchronous request handling.
- **Unified Data Strategy:** All operational data—including user accounts, notes, quiz results, and high-dimensional vector embeddings—shall be stored within a unified PostgreSQL environment. This ensures immediate relational integrity and consistent semantic search capabilities without the overhead of a multi-database architecture.
- **AI Orchestration:** Intelligence features must be powered by LangChain-based orchestration, connecting the application logic to third-party Large Language Models (LLMs) via secure API integrations.
- **External API Dependency:** Since AI services rely on third-party APIs (e.g., OpenAI), the system must implement robust handling for rate limits and cost constraints through backend circuit breakers and request queuing.
- **Cloud Resource Management:** To control processing latency and storage costs, a strict 50 MB file size limit per upload (PDF/PPT) shall be enforced. All files must be offloaded to Cloudinary or AWS S3 to prevent database bloating.
- **Data Privacy & Ethics:** To maintain high security standards, no custom global models shall be trained on user data without explicit consent. The system must strictly adhere to data privacy requirements, ensuring individual study materials remain private and secure.

2.6 User Documentation

Planned documentation:

- **Learner Guide:** How to upload materials, generate notes, take quizzes, and manage tasks.
- **Quick Start Tutorial:** First-time onboarding with examples.
- **FAQ and Troubleshooting:** Common issues (upload failures, AI errors, etc.).
- **Admin Guide:** Managing users, monitoring usage, and handling incidents.

2.7 Assumptions and Dependencies

Assumptions:

- Users have access to reliable internet and a compatible device.
- Users provide legal rights to process the documents they upload.
- External AI and calendar APIs remain available and stable.

Dependencies:

- Availability and performance of external AI APIs and vector store.
- Email/notification providers for sending alerts.
- Cloud database and hosting infrastructure.

3. External Interface Requirements

3.1 User Interface

NeuraNote provides role-specific, responsive GUIs for web, following a consistent design system.

3.1.1 Learner Interface

- Dashboard showing recent notes, and upcoming tasks.
- Screens for:
    - Uploading materials.
    - Viewing and editing notes.
    - Taking quizzes.
    - Managing to-dos and calendar events.
    - Viewing progress charts and history.

3.1.2 Admin Interface

- Overview of user statistics and system status.
- User list with basic management actions (disable account, reset password, etc.).
- Logs or summaries of usage (high-level only; no access to private note content).

3.1.3 GUI Standards and Constraints

- Consistent navigation bar, typography, and button styles across all pages.
- Dark/light mode support is desirable (optional).
- Forms validate user input and show clear error messages.

3.2 Hardware Interfaces

- No special hardware is required beyond standard desktops, laptops
- Server environment must provide sufficient CPU, RAM, and storage for AI requests and indexing.

3.3 Software Interfaces

- **Database Interface:** The system interfaces with a **PostgreSQL** instance to perform CRUD operations on user data and execute semantic similarity searches for note retrieval.

- **AI Orchestration Interface:** The backend utilizes **LangChain** to communicate with third-party LLM APIs (e.g., OpenAI) for generating notes, quizzes, and flashcards.

- **Storage Interface:** The application connects to **Cloudinary/AWS** APIs for the persistent storage and retrieval of uploaded study materials.

- **Authentication Interface:** Secure session management is handled via **JWT (JSON Web Tokens)** issued by the FastAPI backend.
- **External Services:**

    - AI/LLM API for note and quiz generation.
    - Calendar API (e.g., Google Calendar) for synchronization of study events.
    - Email or push-notification providers.

3.4 Communication Interfaces

- All client–server communication over HTTPS using RESTful APIs (JSON).
- Authentication via JWT in the `Authorization` header.
- WebSockets or Server-Sent Events may be used for real-time notifications (optional)

4. System Features

4.1 Login & Authentication

This feature provides secure access to NeuraNote for learners and admins.

4.1.1 Description and Priority

- High priority.
- Ensures that only authorized users can access personal content and system tools.

4.1.2 Stimulus/Response Sequences

1. User opens the login page or app.
2. User enters email and password.
3. System validates credentials and, if correct, generates a JWT and redirects to the dashboard.
4. If credentials are invalid, the system shows an error and allows retry (with a limit).
5. User can request a password reset link via email.

4.1.3 Functional Requirements

- **REQ-4.1.1**: The system must allow users to register using a unique email and password.
- **REQ-4.1.2**: The system must authenticate login attempts and indicate success or failure
- **REQ-4.1.3**: The system must lock further attempts or apply rate limiting after a configurable number of failed logins.
- **REQ-4.1.4**: The system must issue a secure JWT for authenticated sessions.
- **REQ-4.1.5**: The system must provide a password reset mechanism via email.
- **REQ-4.1.6**: The system must support logout, invalidating the active token on the client side.

4.2 Study Material Upload & Processing

This feature lets learners upload and preprocess study content.

4.2.1 Description and Priority

- High priority.
- Acts as the entry point for generating AI-powered notes and quizzes.

4.2.2 Stimulus/Response Sequences

1. Learner navigates to “Upload Materials”.
2. Learner selects one or more files (PDF, DOCX, images, etc.).
3. System validates file types and sizes.
4. System uploads files and displays progress.
5. System extracts text and metadata, storing them for later AI processing.
6. On failure, the system displays an error and allows retry.

4.2.3 Functional Requirements

- **REQ-4.2.1**: The system must support uploading at least PDF, DOCX, TXT, and common image formats.
- **REQ-4.2.2**: The system must validate file size against a configured maximum limit.
- **REQ-4.2.3**: The system must extract textual content from uploaded files where possible.
- **REQ-4.2.4**: The system must store uploaded files and extracted content linked to the learner’s account.
- **REQ-4.2.5**: The system must show upload progress and completion or failure status.
- **REQ-4.2.6**: The system must handle upload failures gracefully and allow re-upload.

4.3 AI-Driven Note Generation

This feature uses AI to convert materials and prompts into structured notes.

4.3.1 Description and Priority

- High priority.
- Central to NeuraNote's value as an intelligent study companion.

4.3.2 Stimulus/Response Sequences

1. Learner selects one or more uploaded materials or an existing note.
2. Learner provides optional instructions (e.g., "summarize for exam", "focus on definitions").
3. System sends relevant content and instructions to the AI service.
4. AI service returns structured notes (headings, bullet points, explanations).
5. System displays the generated draft; learner can accept, edit, or regenerate.

4.3.3 Functional Requirements

- **REQ-4.3.1**: The system must allow learners to select source materials for note generation.
- **REQ-4.3.2**: The system must send content and user instructions to an AI backend for processing.
- **REQ-4.3.3**: The system must receive and store generated notes as editable documents.
- **REQ-4.3.4**: The system must allow learners to edit AI-generated notes and save changes.
- **REQ-4.3.5**: The system must handle AI request failures (timeouts, errors) and present an appropriate message with retry.

4.4 Note Management

This feature allows learners to organize and work with their notes.

4.4.1 Description and Priority

- High priority.
- Enables long-term use of generated notes for learning and revision.

4.4.2 Stimulus/Response Sequences

1. Learner opens the “Notes” section.
2. System lists notes grouped by course/subject or tags.
3. Learner selects a note to view or edit.
4. Learner may rename, retag, or move the note to another collection.
5. Learner can export a note as PDF or text.

4.4.3 Functional Requirements

- **REQ-4.4.1**: The system must display a list of notes with basic metadata (title, subject, last updated).
- **REQ-4.4.2**: The system must allow learners to edit and save note content.
- **REQ-4.4.3**: The system must allow creating, renaming, and deleting note collections (e.g., courses).
- **REQ-4.4.4**: The system must support tagging notes with user-defined tags.
- **REQ-4.4.5**: The system must allow exporting notes in at least one standard format (e.g., PDF).
- **REQ-4.4.6**: The system must confirm before permanently deleting a note.

4.5 Quiz & Active Learning

This feature supports quizzes generated from notes and tracks performance.

4.5.1 Description and Priority

- High priority.
- Implements active learning and self-testing to improve retention.

4.5.2 Stimulus/Response Sequences

1. Learner selects a note or topic and chooses “Generate Quiz”.
2. System creates questions via AI based on the content.
3. Learner starts the quiz and answers each question.
4. System shows immediate feedback and correct answers.
5. At the end, system displays overall score and stores results.

4.5.3 Functional Requirements

- **REQ-4.5.1**: The system must generate quiz questions (e.g., MCQ, short answer) from selected notes.
- **REQ-4.5.2**: The system must present questions one by one with clear options or input fields.
- **REQ-4.5.3**: The system must evaluate answers and show whether they are correct, with explanations where available.
- **REQ-4.5.4**: The system must record quiz attempts with timestamps and scores.
- **REQ-4.5.5**: The system must allow learners to review past quiz attempts and answers.

4.6 Progress Tracking & Analytics

This feature provides learners with insights into their learning activity and performance.

4.6.1 Description and Priority

- Medium to high priority.
- Helps learners stay motivated and identify weak areas.

4.6.2 Stimulus/Response Sequences

1. Learner opens the “Progress” or “Dashboard” view.
2. System retrieves statistics such as number of notes, quizzes taken, scores, and streaks.
3. System displays charts and indicators (e.g., weekly activity, subject-wise performance).

4.6.3 Functional Requirements

- **REQ-4.6.1**: The system must track aggregate statistics like total quizzes completed and average score.
- **REQ-4.6.2**: The system must display visual summaries (charts or graphs) of performance over time.
- **REQ-4.6.3**: The system must track and show study streaks based on daily activity.
- **REQ-4.6.4**: The system must allow learners to filter analytics by subject or time range.

4.7 To-Do List & Calendar Integration

This feature merges study tasks with calendar scheduling.

4.7.1 Description and Priority

- High priority for productivity aspects.
- Supports planning and time management for learning.

4.7.2 Stimulus/Response Sequences

1. Learner creates a new task with title, due date, and optional link to a note.
2. System saves the task and shows it in a to-do list and calendar view.
3. If calendar integration is enabled, system syncs events with the external calendar.
4. Before due time, the system sends a reminder notification.
5. Learner marks the task as completed or updates the due date.

4.7.3 Functional Requirements

- **REQ-4.7.1**: The system must allow learners to create, edit, and delete tasks.
- **REQ-4.7.2**: The system must allow associating tasks with specific notes or topics.
- **REQ-4.7.3**: The system must provide a calendar view showing tasks and study events.
- **REQ-4.7.4**: The system must support optional synchronization with an external calendar account.
- **REQ-4.7.5**: The system must generate reminders before task due times, based on user preferences.

4.8 Semantic Search (Vector-Based Retrieval)

This feature enables the system to understand the conceptual meaning behind user queries rather than relying on exact keyword matching.

4.8.1 Description and Priority

- Priority: Medium (Staged for Phase 3).
- Goal: To allow learners to retrieve relevant information across large volumes of synthesized notes and study materials using natural language queries.

4.8.2 Stimulus/Response Sequences

- Stimulus: The user enters a conceptual query in the search bar (e.g., "Explain the relationship between the two main theories in my physics notes").
- System Action: The system utilizes LangChain orchestration to convert the query into a vector embedding and performs a similarity search within the ChromaDB vector database.
- Response: The system retrieves and displays the most contextually relevant note segments, even if the exact words from the query do not appear in the text.

4.8.3 Functional Requirements

- **REQ-4.8.1**: The system shall convert all finalized AI-generated notes into high-dimensional vector embeddings for indexing.
- **REQ-4.8.2:** The system must integrate a vector database (e.g., ChromaDB) to store and manage these embeddings linked to the user's account.
- **REQ-4.8.3:** The search interface must support natural language processing (NLP) to interpret complex educational queries.
- **REQ-4.8.4:** The system shall return a ranked list of results based on semantic similarity scores.

4.9 Notification & Reminder System

This feature is designed to maintain the learner’s engagement and ensure the "Habit Formation" goals of the platform are met by alerting users to scheduled tasks and note-related activities.

4.9.1 Description and Priority

- Priority: Medium.
- Goal: To support consistent learning habits and ensure timely review of synthesized materials.

4.9.2 Stimulus/Response Sequences

- **Stimulus:** A scheduled study task becomes due in the Calendar View, or a user-initiated modification loop is completed.
- **System Action:** The system generates a notification entry linked to the specific note or task.
- **Channel Delivery:** The learner receives an in-app notification and, based on their settings, push or email alerts.
- **Learner Action:** The learner interacts with the notification to navigate directly to the relevant Folder or Synthesized Note to begin a study session or initiate a Quiz/Flashcard request.

4.8.3 Functional Requirements

- **REQ-4.8.1:** The system must store all notifications securely within the PostgreSQL database (Phase 1) linked to the specific Registered User account.
- **REQ-4.8.2:** The system must provide a centralized Notification Center within the study dashboard to display historical and active alerts.
- **REQ-4.8.3:** The system must support multi-channel delivery, including In-App, Email, and Mobile Push Notifications.
- **REQ-4.8.4:** The system must allow learners to manage their preferences, enabling or disabling specific channels for different types of alerts.
- **REQ-4.8.5:** Upon the completion of an AI-synthesized note, the system shall notify the user that the material is ready for Active Recall (Quizzes or Flashcards.

4.9 Administrative Management

This feature supports basic system administration.

4.9.1 Description and Priority

- Medium priority.
- Required for monitoring and maintaining the platform.

4.9.2 Stimulus/Response Sequences

1. Admin logs into the admin dashboard.
2. System shows high-level metrics (user count, active users, error rates).
3. Admin searches for a specific user and may disable the account if needed.
4. Admin views basic logs (non-content) for troubleshooting.

4.9.3 Functional Requirements

- **REQ-4.9.1**: The system must allow admins to log in with elevated privileges.
- **REQ-4.9.2**: The system must allow admins to view a list of users and basic account status.
- **REQ-4.9.3**: The system must allow admins to deactivate or reactivate user accounts.
- **REQ-4.9.4**: The system must provide basic monitoring metrics (e.g., API error counts).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Critical actions (e.g., login, opening a note, starting a quiz) should complete within 2 seconds under normal load.
- Non-critical actions (e.g., exporting notes, generating large quizzes) should complete within 5 seconds where possible.
- The system should support at least several hundred concurrent users on the initial deployment.

5.2 Safety Requirements

- The system must avoid data loss by using regular automated backups of databases.
- System failures should not corrupt stored notes, quiz results, or user accounts.

5.3 Security Requirements

- All network communication must use HTTPS with modern TLS.
- Passwords must be stored using strong hashing (e.g., bcrypt).
- Access to user data must be enforced using role-based access control.
- Sensitive content must not be exposed to other users or external parties without explicit consent.

5.4 Software Quality Attributes

- **Reliability:** Aim for high uptime during typical study hours; implement robust error handling.

- **Usability:** Interfaces should be simple and consistent, supporting learners with varying tchnical skills..

- **Maintainability:** Code should follow modular architecture (separate frontend, backend, and services).

- **Portability:** The web client should run on all major browsers

- **Testability:** Core features should be covered by automated unit and integration tests.

5.5 Business Rules

- Only authenticated learners can access private notes and quizzes.
- AI-generated content is stored per user and is not shared by default.
- Users may delete their data, and the system must remove or anonymize it accordingly.